U ser M odel

Space and Allocation M odel

U ser A vailability M odel

R equest M odel

M eeting Tim e Suggestion

Rule Engine

A llocation Strategy and Service

Data Access and Control

Backend Application Facade

User

#userId : int
#nam e : String
#em ail : S tring
#status : String
#prim aryUnitNam e : String
#assignedRoleNames : vector<String>

+getId() : int
+getNam e() : String
+getEm ail() : String
+getStatus() : String
+getPrimaryUnitName() : String
+getAssignedRoleNames() : vector<String>
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean

Student

- studentNo : String
- program: String
- yearLevel : int

+getStudentNo() : String
+getProgram () : String
+getYearLevel() : int
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean

Instructor

- title : String
- officeRoom : String

+getTitle() : String
+getOfficeRoom() : String
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean

TeachingAssistant

- assistantType : String

+getAssistantType() : String
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean

Staff

-jobTitle : String

+getJobTitle() : String
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean

Adm inistrator

- adminLevel : String

+getAdm inLevel() : String
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean

UserProfileData

+em ail : String
+status : String
+studentNo : String
+program: String
+yearLevel : int
+ title : S tring
+officeRoom : String
+assistantType : String
+jobTitle : String
+adm inLevel : String
+prim aryUnitNam e : String
+assignedRoleNames : vector<String>

UserFactory

+createUser(userId : int, name : String, roleName : String) : User*
+createUser(userId : int, name : String, roleName : String, profile : UserProfileData) : User*

Role

- roleId : int
- roleName : String
- perm issions : vector<String>

+getId() : int
+getNam e() : String
+getPerm issions() : vector<String>

OrganizationUnit

- unitId : int
- name : String
- unitType : String

+getId() : int
+getNam e() : String
+getUnitType() : String
+getType() : String
Departm ent

+getType() : String

Faculty

+getType() : String

Club

+getType() : String

AdministrativeOffice

+getType() : String

ResearchGroup

+getType() : String

Space

- id : int
- name : String
- capacity : int
- hasProjector : boolean
- hasWhiteboard : boolean
- hasComputers : boolean
- isAvailable : boolean
- building : String

+getId() : int
+getNam e() : String
+getCapacity() : int
+getHasProjector() : boolean
+getHasW hiteboard() : boolean
+getHasComputers() : boolean
+getIsAvailable() : boolean
+setAvailability(availability : boolean) : void
+getBuilding() : String
+hasFeature(feature : String) : boolean
+getType() : String

Classroom

+getType() : String

Laboratory

+getType() : String

MeetingRoom

+getType() : String

TimeSlot

- day : int
- startM inutes : int
- endMinutes : int

+TimeSlot(day : int, startHour : int, endHour : int)
+fromMinutes(day : int, startMinutes : int, endMinutes : int) : TimeSlot
+getD ay() : int
+getStartM inutes() : int
+getEndMinutes() : int
+getStartHour() : int
+getEndHour() : int
+getStartM inute() : int
+getEndM inute() : int
+getStartTim eString() : String
+getEndTimeString() : String
+overlapsW ith(other : TimeSlot) : boolean
+isValid(day : int, startMinutes : int, endMinutes : int) : boolean

Allocation

- id : int
- requestId : int
- space : Space*
- timeSlot : TimeSlot
- assignedParticipants : int
- status : AllocationStatus

+getId() : int
+getRequestId() : int
+getSpace() : Space*
+getTimeSlot() : TimeSlot
+getAssignedParticipants() : int
+getStatus() : AllocationStatus

AllocationStatus

Active

UserBusySlot

- userId : int
- timeSlot : TimeSlot
- reason : String

+UserBusySlot(userId : int, timeSlot : TimeSlot, reason : String)
+getUserId() : int
+getTimeSlot() : TimeSlot
+getReason() : String

UserAvailabilityService

-workingDayStartMinutes : int
-workingDayEndMinutes : int

+UserAvailabilityService() : void
+UserAvailabilityService(workingDayStartMinutes : int, workingDayEndMinutes : int) : void
+isWithinWorkingHours(slot : TimeSlot) : boolean
+isUserBusy(userId : int, slot : TimeSlot, busySlots : vector<UserBusySlot>) : boolean
+isUserAvailable(userId : int, slot : TimeSlot, busySlots : vector<UserBusySlot>) : boolean
+getBusySlotsForUser(userId : int, busySlots : vector<UserBusySlot>) : vector<UserBusySlot>

Request

- requestId : int
- requester : User*
- requestedSpace : Space*
- participantCount : int
- priority : int
- status : RequestStatus
- title : String
- purpose : String
- requiredFeature : String
- requiredBuilding : String
- rejectionReason : String
- lifecycleHistory : vector<String>

+getId() : int
+getRequester() : User*
+getRequestedSpace() : Space*
+getParticipantCount() : int
+getPriority() : int
+getStatus() : RequestStatus
+getTitle() : String
+getPurpose() : String
+getRequiredFeature() : String
+getRequiredBuilding() : String
+getRejectionReason() : String
+getLifecycleHistory() : vector<String>
+addHistoryEvent(event : String) : void
+m arkApproved() : void
+markRejected(reason : String) : void
+isRecurring() : boolean
+getRequestType() : String

OneTimeRequest

- requestedTimeSlot : TimeSlot

+getRequestedTimeSlot() : TimeSlot
+isRecurring() : boolean
+getRequestType() : String

RecurringRequest

- requestedTimeSlots : vector<TimeSlot>

+getRequestedTimeSlots() : vector<TimeSlot>
+isRecurring() : boolean
+getRequestType() : String

ExamRequest

- courseCode : String
- courseName : String
- examType : String
- canSplitAcrossRooms : boolean
- examTimeSlot : TimeSlot

+getCourseCode() : String
+getCourseName() : String
+getExamType() : String
+getCanSplitAcrossRooms() : boolean
+getExamTimeSlot() : TimeSlot
+isRecurring() : boolean
+getRequestType() : String

CommitteeMeetingRequest

- meetingTopic : String
- durationMinutes : int
- preferredTimeSlot : TimeSlot
- requiredParticipantIds : vector<int>
- participantRoles : vector<String>

+getMeetingTopic() : String
+getDurationM inutes() : int
+getPreferredTimeSlot() : TimeSlot
+getRequiredParticipantIds() : vector<int>
+getParticipantRoles() : vector<String>
+addRequiredParticipant(userId : int, participantRole : String) : void
+isRecurring() : boolean
+getRequestType() : String

RequestParticipant

- requestId : int
- userId : int
- participantRole : String

+getRequestId() : int
+getUserId() : int
+getParticipantRole() : String

InvalidRequest

- requestTypeLabel : String
- rawTimeInfo : String

+getRequestTypeLabel() : String
+getRawTimeInfo() : String
+isRecurring() : boolean
+getRequestType() : String

RequestFactory

+createRequest(requestId : int, requestType : String, requester : User*, requestedSpace : Space*, participantCount : int, title : String, purpose : String, requiredFeature : String, requiredBuilding : String, timeData : String, courseCode : String, courseName : String, examType : String, canSplitAcrossRooms : boolean) : Request* +createInvalidRequest(requestId : int, requestType : String, requester : User*, requestedSpace : Space*, participantCount : int, title : String, purpose : String, requiredFeature : String, requiredBuilding : String, timeData : String, rejectionReason : String, courseCode : String, courseName : String, examType : String, canSplitAcrossRooms : boolean) : Request*

RequestStatus

Pending
Approved
Rejected

MeetingTimeSuggestion

- suggestedTimeSlot : TimeSlot
- spaceId : int
- reason : String
- score : int
-timeDistanceMinutes : int
- dayDistance : int
- conflictCount : int

+MeetingTimeSuggestion(suggestedTimeSlot : TimeSlot, spaceId : int, reason : String, score : int, timeDistanceMinutes : int, dayDistance : int, conflictCount : int)
+getTimeSlot() : TimeSlot
+getSpaceId() : int
+getReason() : String
+getScore() : int
+getTimeDistanceMinutes() : int
+getDayDistance() : int
+getConflictCount() : int

MeetingTimeSuggestionService

- userAvailabilityService : UserAvailabilityService

- areParticipantsAvailable(request : CommitteeMeetingRequest, candidateSlot : TimeSlot, userBusySlots : vector<UserBusySlot>) : boolean
- isRequestedSpaceAvailable(requestedSpace : Space*, candidateSlot : TimeSlot, allocations : vector<Allocation>) : boolean
- buildSuggestion(request : CommitteeMeetingRequest, candidateSlot : TimeSlot, requestedSpace : Space*) : MeetingTimeSuggestion
+suggestTimes(request : CommitteeMeetingRequest, requestedSpace : Space*, userBusySlots : vector<UserBusySlot>, allocations : vector<Allocation>, maxSuggestions : int) : vector<MeetingTimeSuggestion>

RuleEvaluationResult

- passed : boolean
- failureReason : String

+isPassed() : boolean
+getFailureReason() : String

IRequestRule

+evaluate(request : Request) : RuleEvaluationResult

CapacityRule

FeatureRule

StatusRule

LocationRule

UserRoleRule

RequestTypeRule

+evaluate(request : Request) : RuleEvaluationResult

ParticipantAvailabilityRule

- users : vector<User*>*

- userBusySlots : vector<UserBusySlot>*
- availabilityService : UserAvailabilityService

+ParticipantAvailabilityRule() : void
+ParticipantAvailabilityRule(users : vector<User*>, userBusySlots : vector<UserBusySlot>) : void
-findUserById(userId : int) : User*
- formatTimeSlot(slot : TimeSlot) : String
+evaluate(request : Request) : RuleEvaluationResult

AvailabilityRule

- isSpaceAvailable(space : Space*, timeSlot : TimeSlot, existingAllocations : vector<Allocation>) : boolean

+check(request : OneTimeRequest, allocations) : boolean
+check(request : RecurringRequest, allocations) : boolean
+check(request : ExamRequest, allocations) : boolean
+check(request : CommitteeMeetingRequest, allocations) : boolean

RuleEngine

- requestRules : vector<IRequestRule>
- availabilityRule : AvailabilityRule

+evaluate(request : OneTimeRequest, allocations) : RuleEvaluationResult
+evaluate(request : RecurringRequest, allocations) : RuleEvaluationResult
+evaluate(request : ExamRequest, allocations) : RuleEvaluationResult
+evaluate(request : CommitteeMeetingRequest, allocations) : RuleEvaluationResult

RuleEngineFacade

- requestTypeRule : RequestTypeRule
- participantAvailabilityRule : ParticipantAvailabilityRule
- capacityRule : CapacityRule
- featureRule : FeatureRule
- statusRule : StatusRule
- locationRule : LocationRule
- userRoleRule : UserRoleRule
- ruleEngine : RuleEngine

+RuleEngineFacade() : void
+RuleEngineFacade(users : vector<User*>, userBusySlots : vector<UserBusySlot>) : void
+evaluateRequest(request : OneTimeRequest, allocations) : RuleEvaluationResult
+evaluateRequest(request : RecurringRequest, allocations) : RuleEvaluationResult
+evaluateRequest(request : ExamRequest, allocations) : RuleEvaluationResult
+evaluateRequest(request : CommitteeMeetingRequest, allocations) : RuleEvaluationResult

IAllocationStrategy

+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean

GreedyAllocationStrategy

+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean

PriorityAllocationStrategy

+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean

MultiRoomExamGreedyStrategy

- greedyStrategy : GreedyAllocationStrategy

- processSplittableExamRequest(request : ExamRequest, spaces : vector<Space*>, allocations : vector<Allocation>) : boolean
+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean

MultiRoomExamBestFitStrategy

- greedyStrategy : GreedyAllocationStrategy

- processSplittableExamRequest(request : ExamRequest, spaces : vector<Space*>, allocations : vector<Allocation>) : boolean
+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean

SharedRoomExamBestFitStrategy

- greedyStrategy : GreedyAllocationStrategy

- processSplittableExamRequest(request : ExamRequest, spaces : vector<Space*>, allocations : vector<Allocation>, requestById : unordered_map<int, Request*>) : boolean
+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean

AllocationStrategyFactory

- greedyStrategy : GreedyAllocationStrategy
- priorityStrategy : PriorityAllocationStrategy
- multiRoomExamGreedyStrategy : MultiRoomExamGreedyStrategy
- multiRoomExamBestFitStrategy : MultiRoomExamBestFitStrategy
- sharedRoomExamBestFitStrategy : SharedRoomExamBestFitStrategy

-AllocationStrategyFactory() : void
+getInstance() : AllocationStrategyFactory
+getStrategy(strategyName : String) : IAllocationStrategy*

AllocationService

- allocations : vector<Allocation>
- userBusySlots : vector<UserBusySlot>
- meetingTimeSuggestionService : MeetingTimeSuggestionService
- ruleEngineFacade : RuleEngineFacade
- allocationStrategy : IAllocationStrategy*

- appendMeetingTimeSuggestions(request : CommitteeMeetingRequest) : void
- appendMeetingTimeSuggestionsForRejectedCommitteeRequests(requests : vector<Request*>) : void
+AllocationService() : void
+AllocationService(strategyName : String) : void
+AllocationService(strategyName : String, users : vector<User*>, userBusySlots : vector<UserBusySlot>) : void
+getAllocations() : vector<Allocation>
+addExistingAllocation(allocation : Allocation) : void
+processRequests(requests : vector<Request*>) : void
+processRequests(requests : vector<Request*>, spaces : vector<Space*>) : void
+processRequest(request : OneTimeRequest) : boolean
+processRequest(request : RecurringRequest) : boolean
+processRequest(request : ExamRequest) : boolean
+processRequest(request : CommitteeMeetingRequest) : boolean
+printAllocations() : void

DataLoader

+loadUsers(filenam e : String) : vector<User*>
+loadSpaces(filename : String) : vector<Space*>
+loadUserBusySlots(filename : String) : vector<UserBusySlot>
+loadRequestParticipants(filename : String) : vector<RequestParticipant>

AllocationW riter

+writeAllocations(filenam e : String, allocations : vector<Allocation>) : void

RequestResultWriter

+writeRequestResults(filename : String, requests : vector<Request*>) : void

SystemData

+users : vector<U ser*>
+spaces : vector<Space*>
+requests : vector<Request*>
+userBusySlots : vector<UserBusySlot>
+requestParticipants : vector<RequestParticipant>

DataController

+loadAllocationStrategyName(configFile : String) : String
+loadAllData(usersFile : String, spacesFile : String, requestsFile : String, userBusySlotsFile : String, requestParticipantsFile : String) : SystemData
+exportAllocations(file : String, allocations : vector<Allocation>) : void
+exportRequestResults(file : String, requests : vector<Request*>) : void
+cleanupData(data : SystemData&) : void

ResourceAllocationSession

- dataController : DataController
- systemData : SystemData
- strategyName : String
- allocationService : unique_ptr<AllocationService>
- loaded : boolean
- allocationProcessed : boolean

- rebuildAllocationService() : void
+ResourceAllocationSession() : void
+~ResourceAllocationSession() : void
+loadStrategyName(configFile : String) : String
+loadSystemData(usersFile : String, spacesFile : String, requestsFile : String, userBusySlotsFile : String, requestParticipantsFile : String) : boolean
+runAllocation() : boolean
+addExistingAllocation(allocation : Allocation) : void
+exportResults(allocationsFile : String, requestResultsFile : String) : void
+printAllocations() : void
+cleanup() : void
+getAllocations() : vector<Allocation>
+getSystemData() : SystemData&
+getStrategyNam e() : String
+isDataLoaded() : boolean
+hasRunAllocation() : boolean

assignedRoleNames

primaryUnitName

participant m appings

creates and ranks

adds ranked suggestion history

loads and exports through

stores loaded data

runs allocation through

exposes allocations